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* Non-normal matrix : a matrix that does not commute with its conjugate transpose

* matrix A is normal if

$$AA^* = A^*A$$

where A^* is conjugate transpose of A.
for real matrices $A^* = A^T$

* ∴ A is non-normal if

$$AA^* \neq A^*A$$

* Normal matrices very nice properties :
  - eigen vectors form orthonormal basis
  - can be diagonalized by a unitary matrix :

$$A = U\Lambda U^T$$

  - their behavior is completely characterized by eigenvalues.

* Examples of normal matrices :
  1. symmetric matrices $A = A^T$
  2. Hermitian matrices $A = A^*$
  3. orthogonal matrices $A^TA = I$
  4. unitary matrices $A^*A = I$

* non-normal matrices can exhibit behavior that eigenvalues alone fail to predict

eg:
$$A = \begin{bmatrix} -1 & 100 \\ 0 & -1 \end{bmatrix}$$

A has eigenvalues $-1, -1$ suggesting stability. But

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots \quad (1)$$

$$A = -I + \underbrace{\begin{bmatrix} 0 & 100 \\ 0 & 0 \end{bmatrix}}_{\text{let } N}$$

$$A = -I + N$$

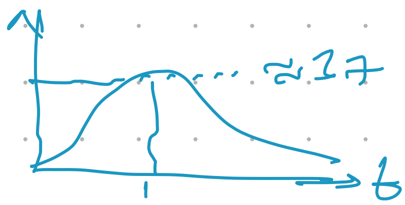
$$e^{At} = e^{-It + Nt}$$
$$= e^{-It} \cdot e^{Nt}$$
$$= e^{-t} e^{Nt}$$

N is nilpotent ($N^2 = 0$, $N^3 = 0 \dots$)

$\therefore$ by (1) $e^{Nt} = I + Nt = \begin{bmatrix} 1 & 100 \\ 0 & 1 \end{bmatrix}$

$$\therefore e^{At} = e^{-t} \begin{bmatrix} 1 & 100 \\ 0 & 1 \end{bmatrix}$$

$$\|e^{At}\| \approx 100\, t e^{-t}$$



transient amplification

* this phenomenon appears in
  - fluid dynamics
  - control theory
  - numerical linear algebra
  - deep learning optimization
  - GSP

Geometric Intuition

* a normal matrix acts independently along orthogonal eigenvector directions

* a non-normal matrix has non-orthogonal eigenvectors. it may not even be diagonalizable.

* $\therefore$
  - different directions interact
  - energy can temporarily grow even when all eigenvalues indicate decay
  - small perturbations can significantly change the spectrum

* for a normal matrix eigenvalue

  singular value $\sigma_i(A) = |\lambda_i(A)|$

* this does not hold for non-normal matrix

* Eigenvalues satisfy $A\underline{v} = \lambda \underline{v}$

* singular values are elements of Σ in

  $A = U \Sigma V^*$

* Eigenvalues describe
  - how matrix acts on eigenvectors
  - asymptotic growth / decay
  - oscillation frequencies
  - long-term dynamics

* singular values describe
  - maximum stretching of vectors

$$\sigma_{max}(A) = \max_{\|x\|=1} \|A x\|$$

  - how much A can amplify signals
  - conditioning
  - transient growth

* fundamental inequality

$$\max_i |\lambda_i| \leq \sigma_{max}(A) \quad \Big| \quad \min_i |\lambda_i| \geq \sigma_{min}(A)$$

  - for normal matrices equality holds
  - for non-normal matrices strict inequality

* for any matrix A

$$\|A\|_2 = \sigma_{max}(A)$$

for invertible A : $\|A^{-1}\|_2 = \dfrac{1}{\sigma_{min}(A)}$